

The unprecedented growth in the Indian electronics demand has spawned a corresponding spurt in the domestic power electronics industry—Dalip Sharma, managing director, Delta-India and SAARC



tifier), DC-to-AC (inverter), DC-to-DC (converter) and AC-to-AC (converter).

While DC-DC converters are used in most mobile devices, AC-DC converters are used when an electronic device (computer, television, etc) is connected to the mains. AC-AC converters are used to change either the voltage level or the frequency (like for international power adaptors and light dimmers). DC-AC converters, on the other hand, are used primarily in UPS or emergency lighting systems.

India's power story

Let's delve a little deeper into India's growth story. Power electronics is very important in the industrial, energy and automotive sectors, among others. As these sectors grow in India, so do the jobs that require training in power electronics. With large investments pouring in from private-sector players like Tata Power, Sterlite Industries (India), Jindal India Thermal Power and Lanco Infratech, power is likely to become one of the largest job-generating sectors in coming years.

According to Dr Bhaskar Bhattacharya, dean-school of engineering, Sharda University, "Many new nuclear power generation plants have been scheduled to start construction. In addition, the manufacturing industry is

picking up steam. So electrical equipment is a new area where power electronics engineers are needed."

He adds, "India is catching up with the West in electrical equipment usage. So for the next few decades, growth will be more than the supply of qualified power electronics engineers."

Power electronics is a dynamic sector. It is becoming popular owing to its presence, usage and overlap with many other sectors. "The industrial sector offers job opportunities in the motor drives area. In the energy

Power electronics is more infrastructure-based than consumer electronics. So there is usually a lot more engineering support, in addition to the engineering design—Dr Bhaskar Bhattacharya, dean-school of engineering, Sharda University



sector, it offers opportunities in companies that work on solar and wind energy conversion and power system compensation. The automotive sector offers job opportunities in the design of various automobile sub-systems, and electric and hybrid vehicles. Many of these are related to energy efficiency and usage. In the present energy scenario, opportunities and prospects for

power electronics engineers are very good," opines Dr Mukul Chandorkar, professor-electrical engineering department, IIT Bombay.

Many multinational companies in the electronics industry have leveraged India's manpower advantage to grow in the domestic market as well as source products and services from India. There is a great demand for highly talented workforce, especially for design and engineering services, with good communication skills in technical and management fields. India's capabilities in IT and engineering make it an attractive destination for sourcing engineering services such as R&D and design.

Power game

The power sector comprises a wide range of firms from fairly large companies to small and medium-scale busi-

Jobs are typically available in the areas of industrial system design, motion control, robotics, efficient energy conversion, and interfaces for solar and wind electricity generation—Dr Mukul Chandorkar, professor-electrical engineering department, IIT Bombay



nesses. Hence this industry requires professionals with varied skillsets to cater to product design and development, project engineering, product management, marketing and sales. The companies in this sector also look for professionals having a superior understanding of electronics, instrumentation, electrical, mechanical and power engineering.

The power sector offers huge opportunities for engineering professionals with multidisciplinary interests such as magnetics and mechanics. Technically satisfying careers can also be pursued in the fields of technical marketing, design engineering, project management, and integration and servicing.

"If we broadly want to define job responsibilities, the different roles